The February 6, 2023 Kahramanmaraş Earthquake and Migration Dynamics

Damage, Displacement, and Recovery in 11 Earthquake-Affected Provinces

Şevval Miray Başar and Büşra Halil

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Introduction

The earthquakes that struck on February 6, 2023, with magnitudes of 7.7 and 7.6 and centered in Kahramanmaraş, are considered one of the largest disasters in Turkey's recent history. The earthquakes in the region caused widespread destruction across a large geographical area. It has not only caused physical destruction but has also led to significant changes in the region's social, economic, and demographic structures. Following the disaster, people were forced to leave their homes, leading to a large-scale internal migration. This study examines 11 provinces affected by the earthquake. The study aims not only to demonstrate the extent of the damage but also to analyze its impact on population and migration patterns.

Within this scope, a study was conducted focusing on the following research questions:

• Which provinces have been the destination for population movements after the earthquake? • Have migration flows concentrated in major cities, or have they been directed toward nearby regions based on geographical proximity? • What is the level of recovery in the earthquake-affected provinces during the period following the earthquake? • Have the provinces managed to return to their expected population trends by 2025?

Data and Preprocessing

In this study, analyses were conducted by combining data from various sources. The datasets used are structured in a way that allows for the analysis of damage, population, and migration flows together. Various preprocessing steps were applied during the data preparation process. For example, all province names were converted to a standard format, operations such as population-weighted ratios and net migration calculations were performed.

These processed data structures were saved in .RData format for future reuse. Download final RData file

The dataset used is shown below:

Table 1: Summary of datasets used in the project.

Dataset	Content	Period	Main_Use
tr_nufus.xlsx	Annual province-level population by sex (TÜİK,ADNKS)	2018–2025	Population trend and recovery analysis

Dataset	Content	Period	Main_Use
deprem_hasar_tablosu.xlsx	Housing damage by province and damage severity (AFAD, ÇŞB)	2023	Damage intensity calculation
deprem_2023_goc_akislari_duzenlenmis.xlsx	Cleaned 2023 migration flows from earthquake provinces (TÜİK)	2023	Migration flow visualization
deprem_goc_ham_filtreli.xlsx	Annual inter-provincial migration data (TÜİK)	2018–2025	Net migration rate analysis
I_ller_Arası_Go_c_.xlsx	Full 81x81 inter-provincial migration table (TÜİK)	2008–2025	Supplementary Ankara analysis

Exploratory Data Analysis

In this section, the first phase of the analysis was conducted. The overall situation was assessed by examining the general trends in the data and addressing the damage and population changes caused by the earthquake.

Damage Intensity

This map illustrates the extent of damage caused by earthquakes in the region across the affected provinces. The 11 earthquake-affected provinces are color-coded based on the number of heavy/collapsed housing per 1,000 people. The darker the color of a province, the more severe the damage relative to its population.

Hatay, which experienced the most severe destruction, is shown in a dark shade, with a heavy/destroyed housing density of 127.7 per 1,000 people. The provinces with the next highest levels of damage are Adıyaman at 88.6, Malatya at 88.0, and Kahramanmaraş at 84.4.

In provinces where the damage is severe, the rate of severe destruction is around 20–25 percent. In other words, approximately one-fourth of these cities has suffered severe destruction.

In provinces such as Diyarbakır, Adana, and Şanlıurfa, the damage is observed to be at a lower rate. The damage has not been distributed equally across all provinces in the region.

Housing Damage Chart

The graph below shows the percentages of homes with light damage, medium damage, and heavy/collapsed damage in the earthquake-affected provinces. The graph is color-coded by damage type, making it possible to observe the damage rates in each province. The total bar length indicates the overall damage intensity for that province. The size of the "heavy/collapsed" section shown in red on the graph indicates that physical destruction has reached a critical level in those provinces.

Accordingly, Hatay, Kahramanmaraş, Malatya, and Adıyaman stand out both in terms of total damage and the intensity of heavy/collapsed damage. The significant rate of heavy damage in Hatay has also had a major impact on migration patterns in that province.

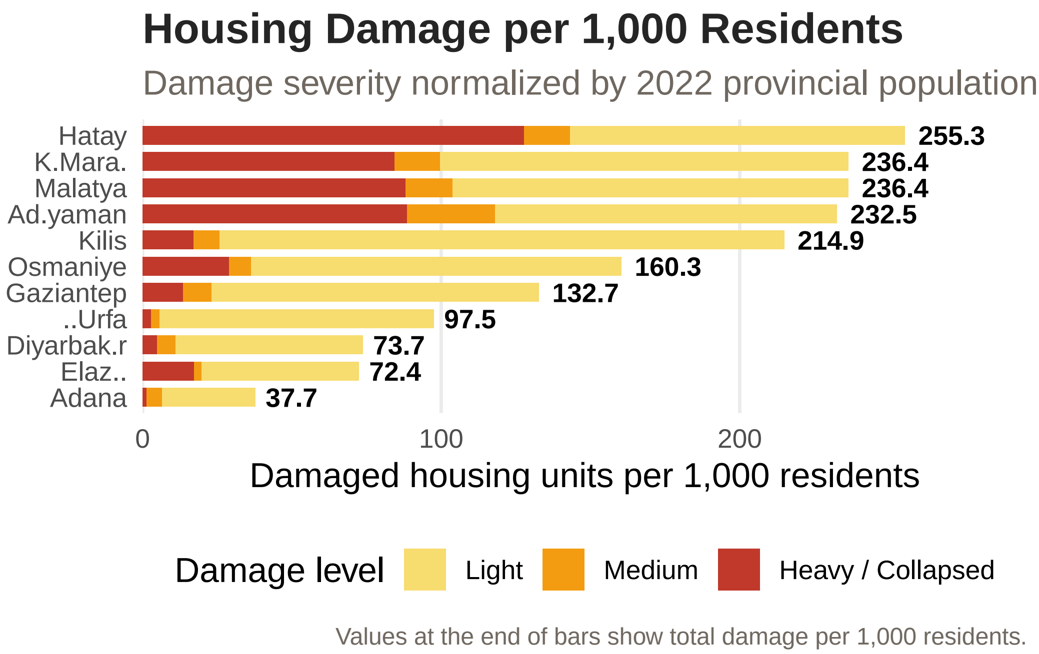

Figure 1: Housing damage intensity by province, normalized by 2022 population.

Migration Overview

The chart below compares the baseline year (2022), the earthquake year (2023), and the first recovery year (2024). An analysis of migration patterns in the earthquake-affected provinces during these years indicates that Malatya, Hatay, Kahramanmaraş, and Adıyaman experienced significant population loss in 2023.

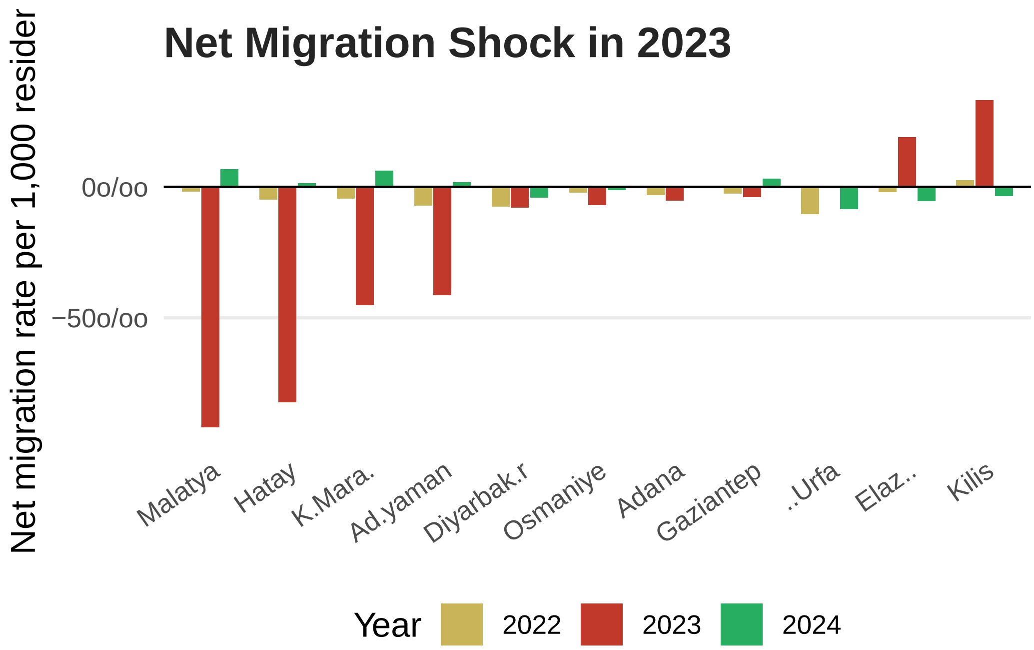

Figure 2: Net migration rate per 1,000 residents in 2022, 2023, and 2024.

Population Trends

The graph below shows the total population change in the 11 earthquake-affected provinces between 2018 and 2025. In 2023, significant population declines were observed particularly in provinces that sustained heavy damage in the earthquake, such as Hatay, Malatya, and Kahramanmaraş. Although signs of recovery were seen in some provinces following the earthquake year, this recovery is not at the same level across all provinces. This graph indicates that the earthquake-affected provinces are still in the process of demographic recovery.

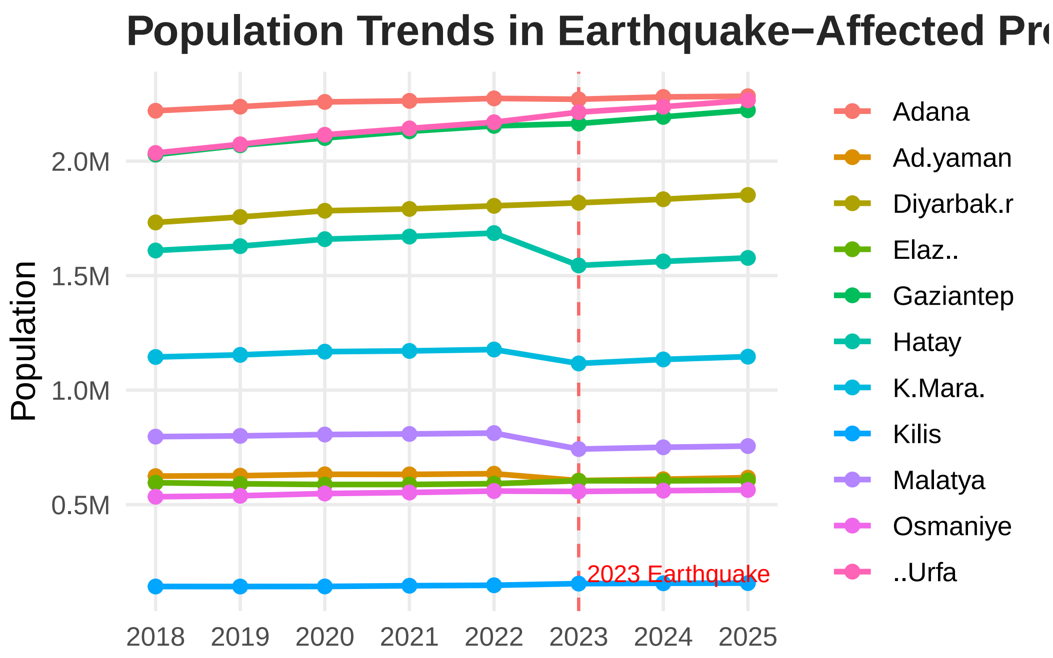

Figure 3: Population trend in 11 earthquake-affected provinces, 2018–2025.

Main Analysis

In this section, detailed analyses were conducted to provide answers to the research questions. The analyses examine the destinations of population flows after the earthquake, changes in the population, and the demographic effects of physical damage.

In this context, migration movements from the earthquake-affected provinces were first examined to determine the destinations of these flows, and the provinces receiving the highest levels of migration were identified. One of the key assessments in the study evaluated the effects of the level of physical damage on population loss.

Finally, the recovery process was examined to assess the degree to which the earthquake-affected provinces have been able to return to expected population trends.

Interactive Migration Flow Map

The map below shows out-migration flows from provinces affected by the 2023 earthquake. By using the filters on the map, you can view out-migration flows from a selected earthquake-affected province. The blue dots on the map, which indicate destination provinces, increase in size based on the volume of migration to that province. It is evident that major cities such as Istanbul, Ankara, Izmir, Antalya, and Mersin serve as significant migration destinations and are highly preferred. Additionally, provinces located in the same geographical area as earthquake-affected regions, such as Konya, Kayseri, and Mardin, have also become important destinations.

```
coords <- tribble(
  ~province, ~lat, ~lon,
  "ADANA",37.00,35.32, "ADIYAMAN",37.76,38.28,
  "DIYARBAKIR",37.91,40.22, "ELAZIG",38.67,39.22,
  "GAZIANTEP",37.07,37.38, "HATAY",36.20,36.16,
  "KAHRAMANMARAS",37.58,36.92, "KILIS",36.72,37.12,
  "MALATYA",38.35,38.31, "OSMANIYE",37.07,36.25,
  "SANLIURFA",37.16,38.80, "ANKARA",39.93,32.86,
  "ISTANBUL",41.02,28.97, "MERSIN",36.80,34.64,
  "ANTALYA",36.90,30.70, "IZMIR",38.42,27.14,
  "KONYA",37.87,32.48, "KAYSERI",38.72,35.49,
  "MARDIN",37.31,40.74, "BURSA",40.18,29.06,
  "KOCAELI",40.77,29.95, "MUGLA",37.21,28.36,
  "AYDIN",37.85,27.84, "ESKISEHIR",39.77,30.52,
  "SAMSUN",41.29,36.33, "SIVAS",39.75,37.02
)

labels <- tribble(
  ~province, ~label,
  "ADANA","Adana", "ADIYAMAN","Adıyaman",
  "DIYARBAKIR","Diyarbakır", "ELAZIG","Elazığ",
  "GAZIANTEP","Gaziantep", "HATAY","Hatay",
  "KAHRAMANMARAS","K.Maraş", "KILIS","Kilis",
  "MALATYA","Malatya", "OSMANIYE","Osmaniye",
  "SANLIURFA","Ş.Urfa", "ANKARA","Ankara",
  "ISTANBUL","İstanbul", "MERSIN","Mersin",
  "ANTALYA","Antalya", "IZMIR","İzmir",
  "KONYA","Konya", "KAYSERI","Kayseri",
  "MARDIN","Mardin", "BURSA","Bursa",
  "KOCAELI","Kocaeli", "MUGLA","Muğla",
  "AYDIN","Aydın", "ESKISEHIR","Eskişehir",
  "SAMSUN","Samsun", "SIVAS","Sivas"
)

flows_map <- flows_2023 |>
  filter(value >= 500, to %in% coords$province) |>
  group_by(from_key = from, to_key = to) |>
  summarise(value = sum(value), .groups = "drop")

tr_simple <- tr_map |>
  st_transform(4326) |>
  st_simplify(dTolerance = 0.03) |>
  left_join(damage |> select(province, damage = heavy_per_1000), by = "province")

map_geojson <- geojsonsf::sf_geojson(tr_simple)
flow_json <- jsonlite::toJSON(flows_map, auto_unbox = TRUE)
coord_json <- jsonlite::toJSON(coords, auto_unbox = TRUE)
label_json <- jsonlite::toJSON(labels, auto_unbox = TRUE)

button_list <- c("ALL", EQ_KEYS)
button_labels <- c("All", labels$label[match(EQ_KEYS, labels$province)])
```

```
buttons_html <- paste0(
  paste0(
    "<button class='prov-btn", ifelse(button_list == "ALL", " active", ""),
    "' onclick=\"setFilter('", button_list, "', this)\">",
    button_labels, "</button>"
  ),
  collapse = ""
)

HTML(glue::glue('
<style>
#controls {{ display:flex; gap:8px; flex-wrap:wrap; align-items:center; margin-bottom:8px; fon
.prov-btn {{ font-size:11px; padding:4px 9px; border-radius:18px; cursor:pointer; border:1px s
.prov-btn.active {{ border-color:#8b1e1e; background:#fdecea; color:#8b1e1e; font-weight:bold;
#tooltip {{ position:fixed; background:white; border:1px solid #ccc; border-radius:8px; paddin
#legend {{ display:flex; gap:16px; font-size:12px; margin-top:8px; flex-wrap:wrap; }}
.leg-item {{ display:flex; align-items:center; gap:5px; }}
.leg-dot {{ width:11px; height:11px; border-radius:50%; }}
</style>
<div id="tooltip"></div>
<div id="controls"><b>Filter by origin:</b>{buttons_html}</div>
<svg id="mapsvg" width="100%" viewBox="0 0 680 420" style="display:block">
  <g id="map-layer"></g><g id="flows-layer"></g><g id="dots-layer"></g><g id="labels-layer"></
</svg>
<div id="legend">
  <div class="leg-item"><div class="leg-dot" style="background:#f0d27a"></div><span>Low damage
  <div class="leg-item"><div class="leg-dot" style="background:#e17035"></div><span>Medium dam
  <div class="leg-item"><div class="leg-dot" style="background:#4b000f"></div><span>High damag
  <div class="leg-item"><div class="leg-dot" style="background:#1a5276"></div><span>Destinatio
</div>
<script>
const TURKEY = {map_geojson};
const FLOWS = {flow_json};
const COORDS_RAW = {coord_json};
const LABELS_RAW = {label_json};
const COORDS = Object.fromEntries(COORDS_RAW.map(d => [d.province, [d.lat, d.lon]]));
const LABELS = Object.fromEntries(LABELS_RAW.map(d => [d.province, d.label]));
const LON_MIN=25.5, LON_MAX=45.2, LAT_MIN=35.4, LAT_MAX=42.6;
const SVG_X_MIN=35, SVG_X_MAX=610, SVG_Y_MIN=38, SVG_Y_MAX=315;

function toSVG(lat, lon) {{
  const x = SVG_X_MIN + (lon - LON_MIN) / (LON_MAX - LON_MIN) * (SVG_X_MAX - SVG_X_MIN);
  const y = SVG_Y_MAX - (lat - LAT_MIN) / (LAT_MAX - LAT_MIN) * (SVG_Y_MAX - SVG_Y_MIN);
  return [x, y];
}}

function damageColor(v) {{
  if (v == null || isNaN(v)) return "#d6d6d6";
  if (v < 10) return "#f0d27a";
  if (v < 30) return "#e69f45";
  if (v < 60) return "#d95f0e";
  if (v < 90) return "#b30000";
```

```
  return "#4b000f";
}}

function drawMap() {{
  const ml = document.getElementById("map-layer");
  TURKEY.features.forEach(f => {{
    const damage = f.properties.damage;
    const coords = f.geometry.coordinates;
    const polys = f.geometry.type === "MultiPolygon" ? coords : [coords];
    polys.forEach(poly => {{ poly.forEach(ring => {{
      let d = "";
      ring.forEach((pt, i) => {{
        const [x, y] = toSVG(pt[1], pt[0]);
        d += (i === 0 ? "M" : "L") + x.toFixed(1) + " " + y.toFixed(1) + " ";
      }});
      d += "Z";
      const path = document.createElementNS("http://www.w3.org/2000/svg","path");
      path.setAttribute("d", d);
      path.setAttribute("fill", damageColor(damage));
      path.setAttribute("fill-opacity", damage == null ? "0.48" : "0.88");
      path.setAttribute("stroke", "#ffffff");
      path.setAttribute("stroke-width", "0.7");
      ml.appendChild(path);
    }}); }});
  }});
}}

function dotR(v, maxDest) {{ return 5 + (v / maxDest) * 13; }}

function curvePath(x1, y1, x2, y2) {{
  const mx=(x1+x2)/2, my=(y1+y2)/2;
  const dx=x2-x1, dy=y2-y1, len=Math.sqrt(dx*dx+dy*dy);
  const bulge=Math.min(len*0.22,45);
  const cx=mx + (-dy/len)*bulge;
  const cy=my + ( dx/len)*bulge;
  return `M ${{x1.toFixed(1)}} ${{y1.toFixed(1)}} Q ${{cx.toFixed(1)}} ${{cy.toFixed(1)}} ${{x
}}

let activeFilter = "ALL";

function setFilter(p, btn) {{
  activeFilter = p;
  document.querySelectorAll(".prov-btn").forEach(b => b.classList.remove("active"));
  btn.classList.add("active");
  render();
}}

const tooltip = document.getElementById("tooltip");

function showTip(e, html) {{ tooltip.innerHTML = html; tooltip.style.opacity = 1; moveTip(e);
function moveTip(e) {{ tooltip.style.left = (e.clientX + 14) + "px"; tooltip.style.top = (e.cl
function hideTip() {{ tooltip.style.opacity = 0; }}
```

```
function render() {{
  const fl = document.getElementById("flows-layer");
  const dl = document.getElementById("dots-layer");
  const ll = document.getElementById("labels-layer");
  fl.innerHTML = ""; dl.innerHTML = ""; ll.innerHTML = "";

  const visible = activeFilter === "ALL" ? FLOWS : FLOWS.filter(f => f.from_key === activeFilt
  const destTotals = {{}};

  visible.forEach(f => destTotals[f.to_key] = (destTotals[f.to_key] || 0) + f.value);
  const maxDest = Math.max(...Object.values(destTotals), 1);

  visible.forEach(f => {{
    if (!COORDS[f.from_key] || !COORDS[f.to_key]) return;
    const [lat1, lon1] = COORDS[f.from_key];
    const [lat2, lon2] = COORDS[f.to_key];
    const [x1, y1] = toSVG(lat1, lon1);
    const [x2, y2] = toSVG(lat2, lon2);

    const path = document.createElementNS("http://www.w3.org/2000/svg","path");
    path.setAttribute("d", curvePath(x1,y1,x2,y2));
    path.setAttribute("fill", "none");
    path.setAttribute("stroke", "#8b1e1e");
    path.setAttribute("stroke-width", "1.8");
    path.setAttribute("stroke-opacity", "0.52");
    path.setAttribute("stroke-linecap", "round");
    path.addEventListener("mouseenter", e => showTip(e, `<b>${{LABELS[f.from_key]}} → ${{LABEL
    path.addEventListener("mousemove", moveTip);
    path.addEventListener("mouseleave", hideTip);
    fl.appendChild(path);
  }});

  const allProvs = new Set();
  visible.forEach(f => {{ allProvs.add(f.from_key); allProvs.add(f.to_key); }});

  allProvs.forEach(p => {{
    if (!COORDS[p]) return;
    const [lat, lon] = COORDS[p];
    const [x, y] = toSVG(lat, lon);
    const isDest = destTotals[p] > 0;

    if (isDest) {{
      const dot = document.createElementNS("http://www.w3.org/2000/svg","circle");
      dot.setAttribute("cx", x); dot.setAttribute("cy", y);
      dot.setAttribute("r", dotR(destTotals[p], maxDest));
      dot.setAttribute("fill", "#1a5276");
      dot.setAttribute("opacity", "0.80");
      dot.setAttribute("stroke", "white");
      dot.setAttribute("stroke-width", "1.5");
      dot.addEventListener("mouseenter", e => showTip(e, `<b>${{LABELS[p]}}</b><br>${{Number(d
      dot.addEventListener("mousemove", moveTip);
      dot.addEventListener("mouseleave", hideTip);
      dl.appendChild(dot);
```

```
    }}

    const text = document.createElementNS("http://www.w3.org/2000/svg","text");
    text.setAttribute("x", x + 7);
    text.setAttribute("y", y - 7);
    text.setAttribute("font-size", "10");
    text.setAttribute("font-weight", "700");
    text.setAttribute("fill", "#111");
    text.setAttribute("stroke", "white");
    text.setAttribute("stroke-width", "2.8");
    text.setAttribute("paint-order", "stroke");
    text.textContent = LABELS[p] || p;
    ll.appendChild(text);
  }});
}}

drawMap();
render();
</script>
'))
```

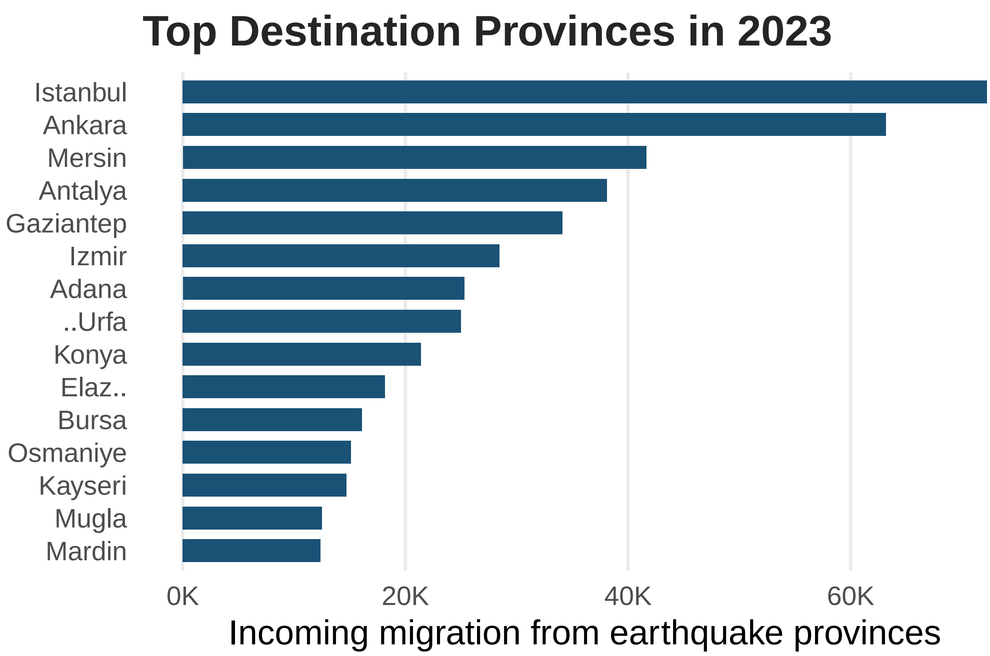

Figure 4: Top destination provinces receiving migrants from earthquake provinces in 2023.

Top Receiving Provinces

This chart shows the ranking of the provinces that receive the most migrants from earthquake-affected areas. Major cities such as Istanbul, Ankara, and Izmir stand out as the top destinations for migrants.

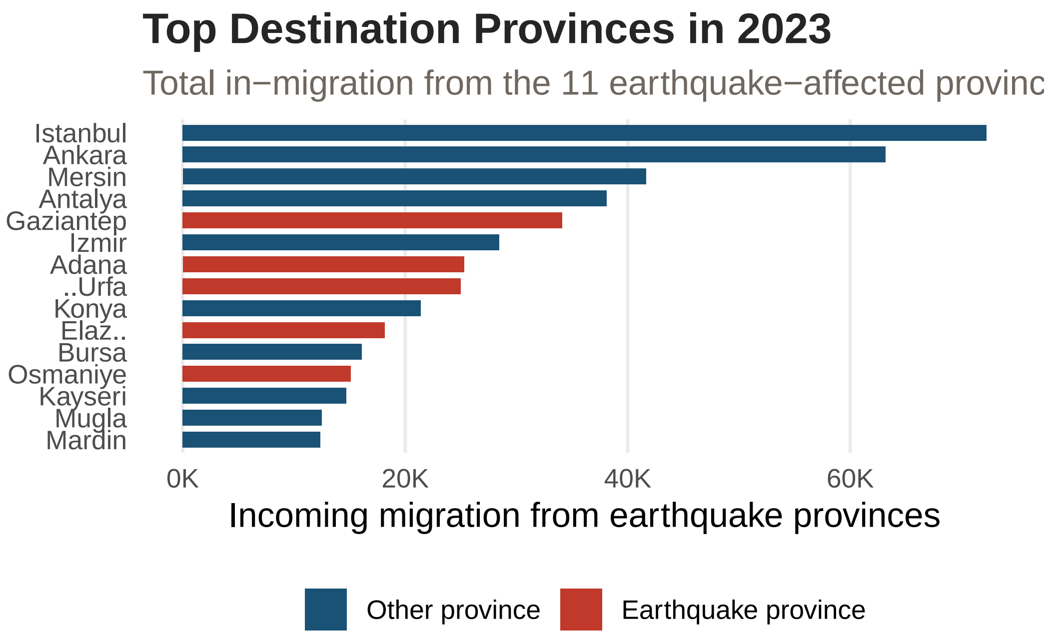

Figure 5: Top destination provinces receiving migrants from earthquake provinces in 2023.

Migration Routes from the Three Hardest-Hit Provinces

Hatay, Malatya, and Kahramanmaraş are among the provinces hardest hit by the earthquake in terms of both the intensity of severe damage and absolute population loss. Since these three provinces constitute a large part of the recorded migration flows in 2023, they were selected as focal points in determining the target provinces and comparing migration routes.

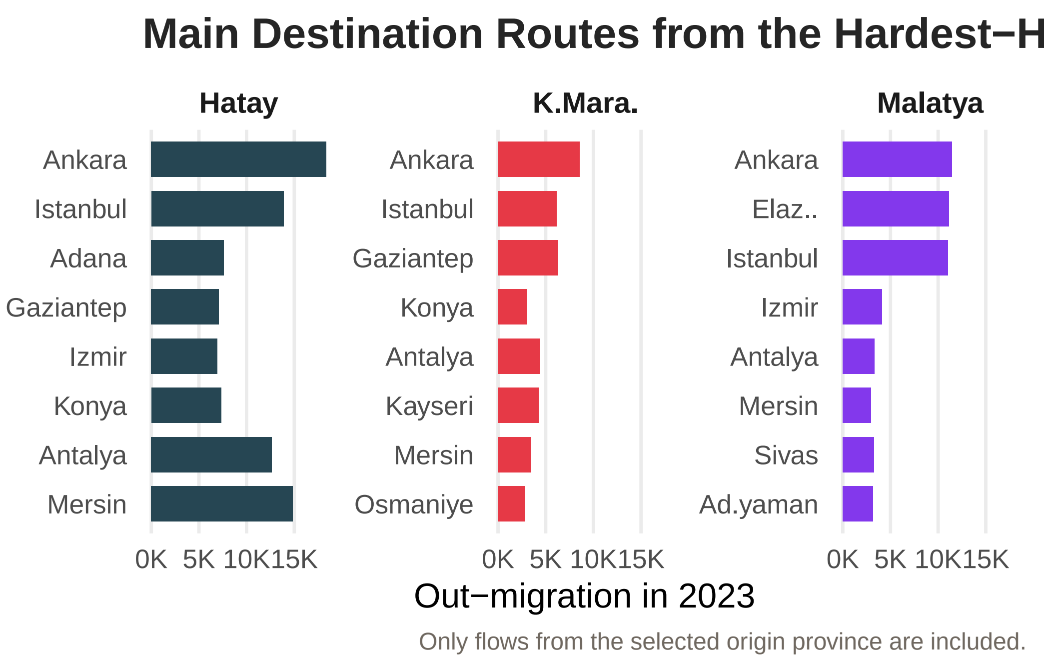

Figure 6: Top destination provinces for out-migrants from Hatay, Malatya and Kahramanmaraş in 2023.

Damage Intensity and Population Loss

This graph addresses the question of whether there is a measurable relationship between physical destruction and demographic loss. This scatter plot, where each point shows the number of severely damaged/collapsed houses per 1,000 people on the horizontal axis and the percentage change in population between 2022 and 2023 on the vertical axis, indicates that as the intensity of severe damage increases, population losses also deepen.

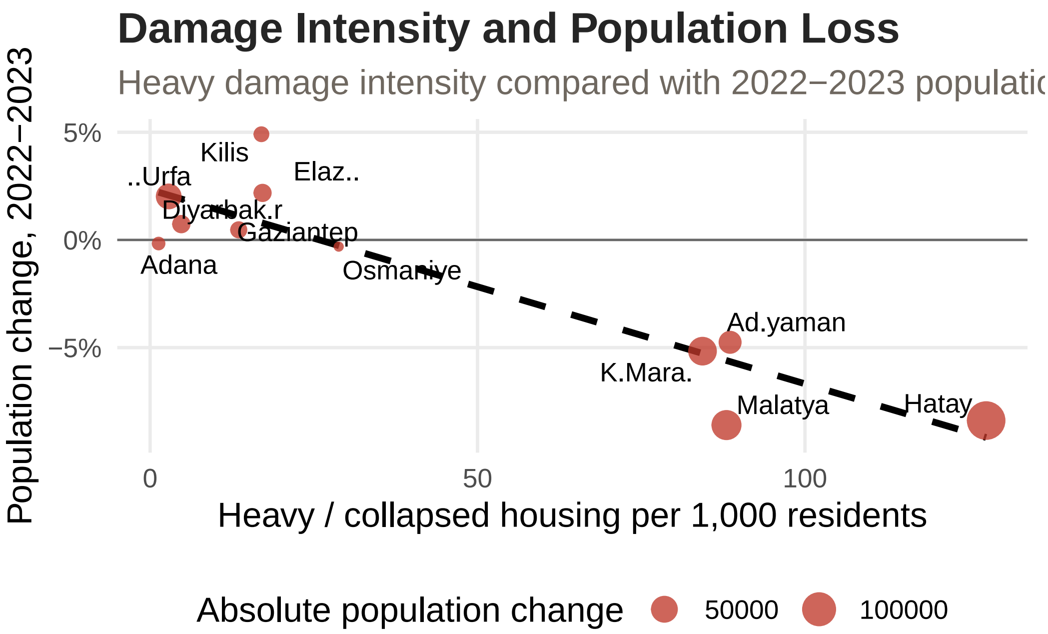

Figure 7: Relationship between heavy damage intensity and population change from 2022 to 2023.

Recovery Status by 2025

The recovery gap shows the extent to which the actual 2025 population of each province deviates from the expected population based on the pre-earthquake (2018–2022) trend. Positive values indicate that the expected trend has been exceeded, while negative values indicate that a persistent demographic gap continues.

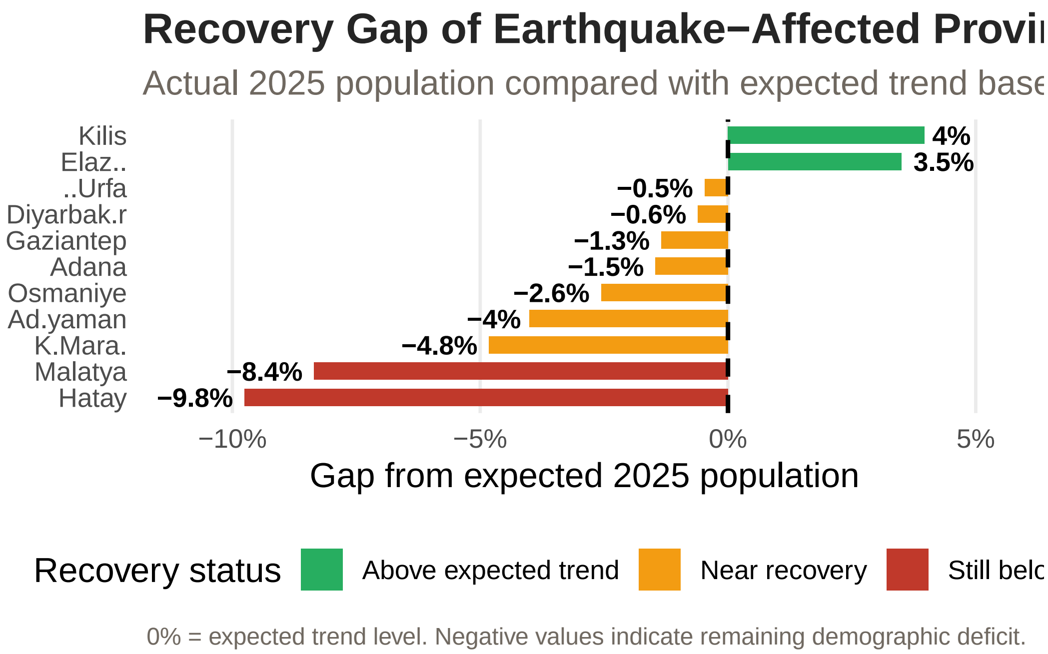

Figure 8: Recovery gap in 2025 relative to expected population based on 2018–2022 trend.

Supplementary Analysis

Ankara Migration Question

Ankara is examined in a separate section as it is one of the major cities to which migration from earthquake zones is most directed. Since looking only at net migration figures can be misleading, in this analysis, incoming and outgoing flows are evaluated together; the component from earthquake provinces and the component from other provinces are separated, and the structural change in Ankara's migration dynamics is revealed more clearly.

The chart below breaks Ankara's net migration into two components to isolate how much of the 2023 surge came specifically from earthquake-affected provinces, as opposed to broader population movements already underway.

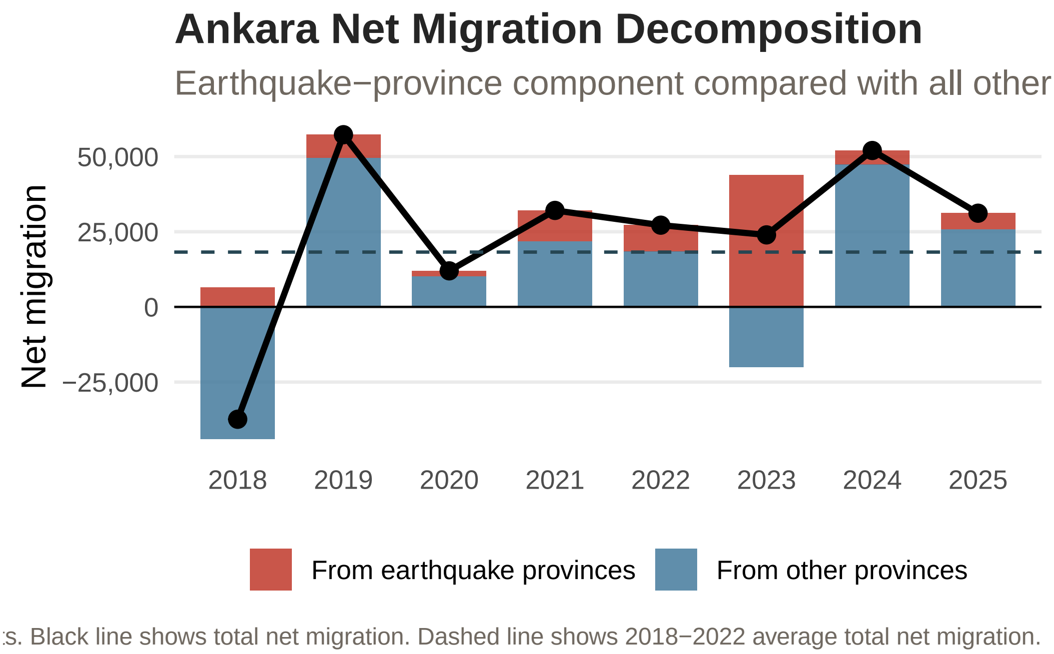

Figure 9: Ankara's net migration from earthquake provinces and other provinces, 2018–2025.

The chart below shows gross in- and out-migration separately, as net figures alone do not distinguish between more arrivals and fewer departures.

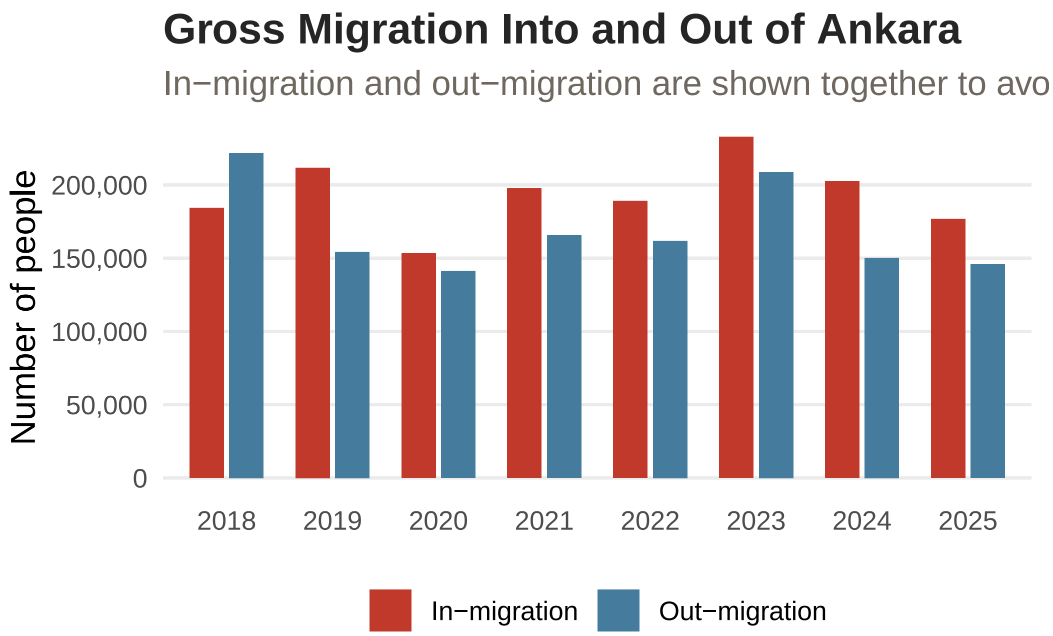

Figure 10: Gross in-migration and out-migration into/from Ankara, 2018–2025.

The table below shows how in-migration to Ankara in 2023 was distributed across the eleven earthquake-affected provinces.

Table 2: Migration from each earthquake province into Ankara in 2023.

Province	In-Migration	Out-Migration	Net Migration	Share of EQ inflow (%)
Hatay	18356	2153	16203	29.0
Malatya	11470	1649	9821	18.2
K.Maraş	8564	1519	7045	13.6
Adıyaman	4136	974	3162	6.5
Adana	5508	2763	2745	8.7
Gaziantep	4197	2536	1661	6.6
Diyarbakır	3588	2476	1112	5.7
Osmaniye	2026	982	1044	3.2
Ş.Urfa	3299	2604	695	5.2
Elazığ	1764	1235	529	2.8
Kilis	284	339	-55	0.4

Key Takeaways

1. Damage intensity was not evenly distributed across the earthquake region; Hatay, Malatya, Adıyaman, and Kahramanmaraş stood out with the highest heavy-damage burden per 1,000 residents.
2. The strongest population losses in 2023 were observed in provinces with higher heavy-damage intensity, suggesting a visible relationship between physical destruction and demographic decline.
3. Migration flows from earthquake-affected provinces were directed not only toward major metropolitan areas such as Istanbul, Ankara, and Antalya, but also toward nearby regional destinations.
4. Recovery by 2025 remained uneven; some provinces approached or exceeded their expected population trend, while the hardest-hit provinces still showed a demographic gap.
5. The Ankara-specific analysis shows that earthquake-origin inflows increased in 2023, but total net migration should be interpreted together with outflows and other provincial movements.

Conclusion

The February 2023 earthquakes left both immediate and lasting demographic traces in the affected provinces. Findings show that the intensity of severe damage is significantly related to population loss; the provinces that experienced the highest destruction also faced the sharpest waves of migration. These migrations in 2023 were directed primarily to large cities, including Istanbul, Ankara, and Antalya, as well as to neighboring regional centers.

As of 2025, the recovery process is not yet complete, and significant differences are observed between provinces. While some provinces have approached or exceeded their pre-earthquake demographic trends, the provinces that suffered the heaviest destruction continue to lag behind expected population levels. When migration flows to Ankara are examined, it is seen that inflows from earthquake zones increased significantly in 2023, but the total net migration continues to be shaped by movements from other provinces.

References

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